

from misuse, unauthorised access, modification, malfunction, destruction or any improper disclosure. Implementation of preventive measures allows users, computers and programs to perform all their permitted critical functions within a secure environment. Network security includes the protection of not just software technologies but also the associated hardware.

Effective network security manages access across the entire network. It targets different types of threats and prevents them from entering or spreading on the network. Network security includes multiple layers of defence at the edge as well as within the network. Each of the network security layers implements its own policies and controls. Only authorised users gain access to the network's resources and malicious actors are blocked from carrying out any exploits or threats. Network security is implemented for both public and private computer networks that are used for everyday jobs like communicating with colleagues in a business or with government agencies, conducting transactions, etc. There are private networks like those within a company and others which might be open for the public. The most common way of protecting a network resource is by assigning a unique name and a corresponding password for it.

Securing a network also involves a complex combination of various hardware devices, such as firewalls, routers, and anti-malware software applications. Businesses and government agencies employ highly skilled information security analysts to implement various security plans and constantly monitor the efficacy of these plans.

The different ways of ensuring network security are:

1. Access control
2. Use of antivirus and anti-malware software
3. Application security
4. Behavioural analytics
5. Prevention of data loss
6. Email security
7. Implementing firewalls
8. Intrusion prevention systems
9. Network segmentation
10. Security information and event management

Listed below are some top vulnerabilities in a network, and the key techniques to protect networks against them.

USB thumb drives

USB drives are one of the most common ways a network can be infected from inside a firewall. There are a couple of reasons for this – these drives are small, inexpensive, hold a lot of data and can also be used on multiple types of computers. The ubiquity of thumb drives has driven hackers to develop a targeted malware, such as the notorious Conficker worm. This malware automatically executes upon connecting with any live USB port. The worst part is that the default OS configurations allow most programs (including the malicious ones) to run automatically. This is like everyone in your neighbourhood having a key to open their own electric

garage door, and being able to use the same key to open the garage doors of everyone else.

Preventive measure

The default auto-run policies of a computer need to be changed.

Wireless access points

Wireless access points provide immediate connectivity to any of the users in the proximity of the network. Wireless attacks by war drivers (the people in vehicles looking for unsecured Wi-Fi networks) are common and have caused significant damage in the recent past. Wireless access points are insecure, regardless of the encryption used. Different protocols like the Wireless Encryption Protocol contain many known vulnerabilities that can be easily compromised with attack frameworks like Aircrack.

Preventive measures

1. WPA2 Enterprise using RADIUS is widely recommended along with an access point that's capable of performing the required authentication and enforcing all security measures.
2. Strong passwords should be used and be changed on a frequent basis.

Laptops and netbooks

Laptops are portable and include full operating systems. They come with a handy Ethernet port that's used for tapping directly into a network. A notebook may already have some malicious code running in the background to scour the network and find some additional systems to infect. Apart from the infected laptops compromising our internal network, it is more important to think about the laptops running the malicious code. All of us have some form of sensitive information that cannot leave the perimeter of the office (like home addresses, account details, medical records, phone numbers, etc.). So when the laptop is attacked, all of this is at risk.

Preventive measures

1. Different open source solutions like TrueCrypt can reduce such risks.
2. Implementation of encrypted file systems for all sensitive data.
3. We should have control over the endpoints that enter and leave the internal system.
4. Different sensitive information like DV, VPN and Wi-Fi access passwords should never be stored persistently on devices like laptops or netbooks.

Miscellaneous USB devices

Thumb drives are not the only USB-connected devices that the IT department needs to be wary of. Many other devices are also capable of storing data on some common file system